

BABU BANARASI **DAS** **UNIVERSITY**

(2026-2027)



SCHOOL OF COMPUTER APPLICATIONS

Project File On
NO SQL & DBaas
(BCADSN23202)

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Bca Ds 27

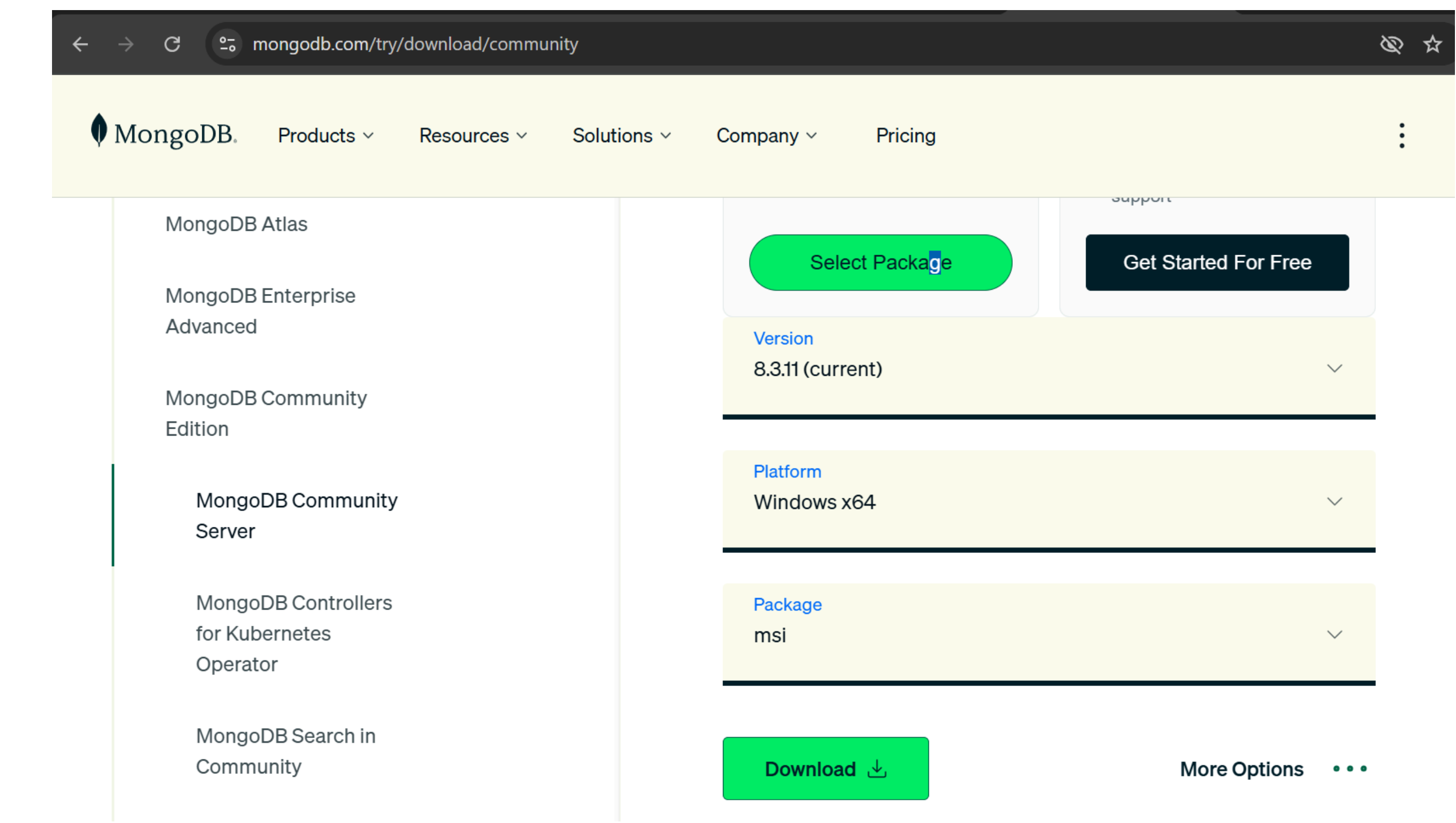
Roll No: 18
(1225258310)

1. Installation and Configuration of MongoDB

MongoDB is a NoSQL, document-oriented database. It stores data in JSON-like BSON documents instead of rows and columns.

Installation steps on Windows:

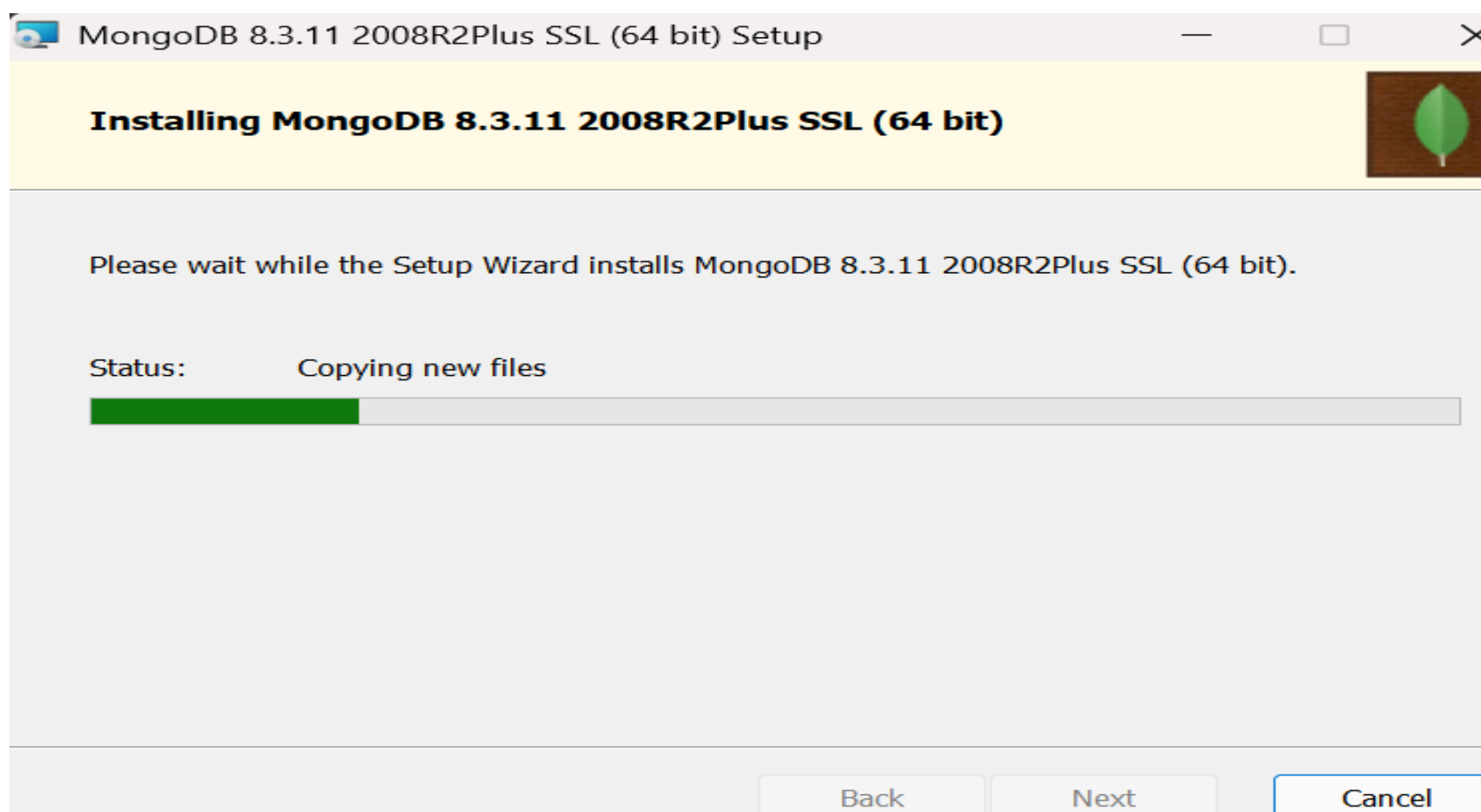
A. Download MongoDB Community Server from the official MongoDB website.



B. Run the installer and select Complete installation.

C. Keep MongoDB Server and MongoDB Compass selected.

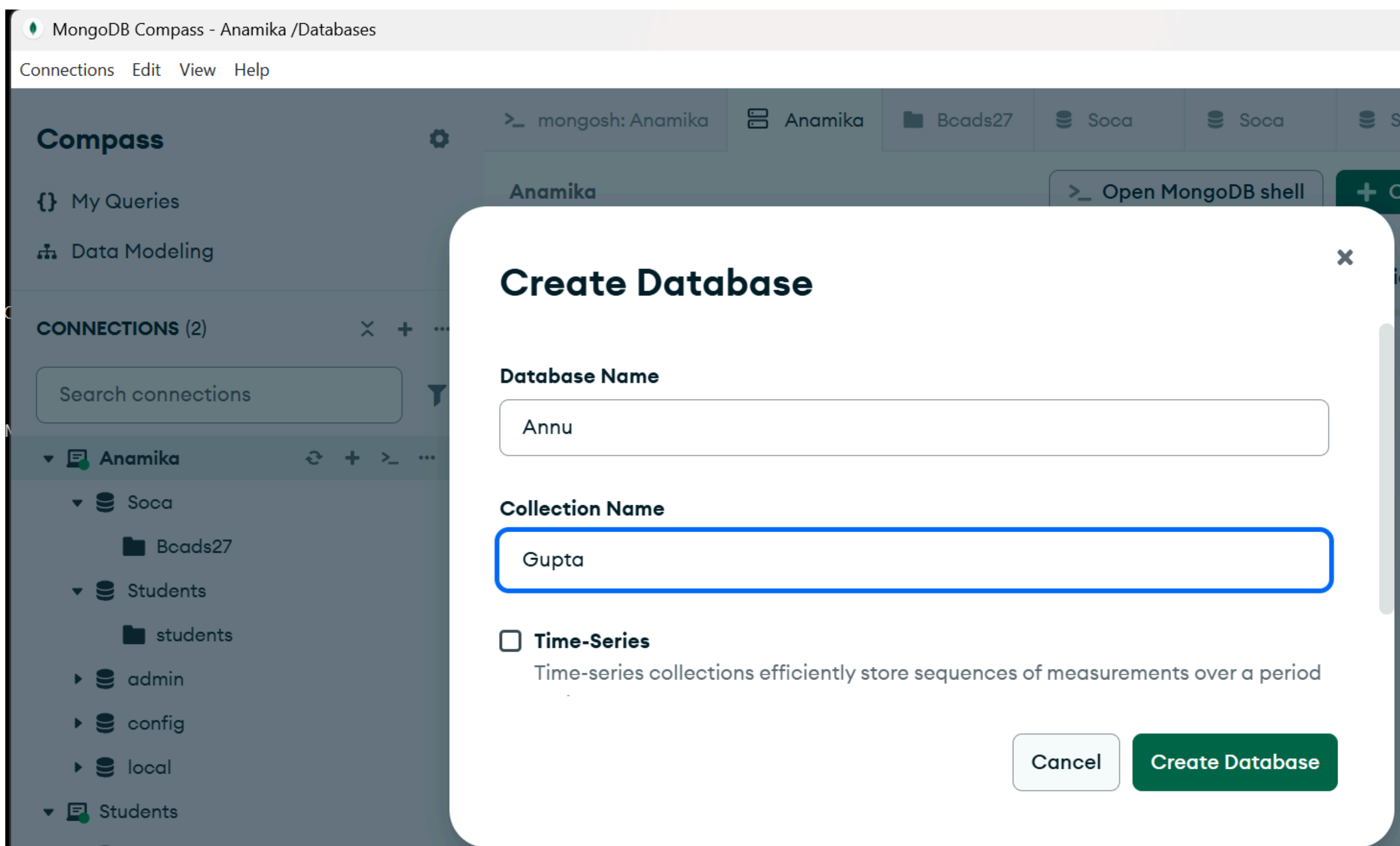
D. Complete the installation.



6. Open MongoDB Compass or MongoDB Shell (mongosh) to work with MongoDB.

2. Working with MongoDB Databases

A. Create/select a database:



B. Check the current database:

```
> _MONGOSH
```

```
> use("Annu")
```

```
< switched to db Annu
```

```
> db.students.insertOne({
```

3. Inserting Documents

A. Insert one document:

```
> use("Annu")
```

```
< switched to db Annu
```

```
> db.students.insertOne({
```

```
  rollNo:101,
```

```
  name:"Sejal",
```

```
  age:20,
```

```
  department:"CSE",
```

```
  marks:85
```

```
})
```

```
< {
```

```
  acknowledged: true,
```

```
  insertedId: ObjectId('6ab0008d167b65c1cd41f650')
```

```
}
```

B. Insert multiple documents:

```

> db.students.insertMany([
  {rollNo:102,name:"Shruti",age:21,department:"ECE",marks:90
},
{rollNo:103,name:"Amit",age:19,department:"ME",marks:75
},
{rollNo:104,name:"Neha",age:22,department:"CSE",marks:88
},
{rollNo:105,name:"Rohit",age:21,department:"CSE",marks:78
}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ab001ce167b65c1cd41f651'),
    '1': ObjectId('6ab001ce167b65c1cd41f652'),
    '2': ObjectId('6ab001ce167b65c1cd41f653'),
    '3': ObjectId('6ab001ce167b65c1cd41f654')
  }
}

```

4. MongoDB Query Operators

Comparison operators:

Operator	Meaning
\$eq	Equal to
\$ne	Not equal to
\$gt	Greater than
\$gte	Greater than or equal to
\$lt	Less than

A. \$gt – Marks greater than 80:

```
> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('6ab0008d167b65c1cd41f650'),
  rollNo: 101,
  name: 'Sejal',
  age: 20,
  department: 'CSE',
  marks: 85
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f651'),
  rollNo: 102,
  name: 'Shruti',
  age: 21,
  department: 'ECE',
  marks: 90
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f653'),
  rollNo: 104,
  name: 'Neha',
  age: 22,
  department: 'CSE',
  marks: 88
}
Annu >
```

B. \$gte – Marks greater than or equal to 85:

```
> db.students.find({ marks: { $gte: 85 } })
< {
  _id: ObjectId('6ab0008d167b65c1cd41f650'),
  rollNo: 101,
  name: 'Sejal',
  age: 20,
  department: 'CSE',
  marks: 85
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f651'),
  rollNo: 102,
  name: 'Shruti',
  age: 21,
  department: 'ECE',
  marks: 90
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f653'),
  rollNo: 104,
  name: 'Neha',
  age: 22,
  department: 'CSE',
  marks: 88
}
```

C. \$lt – Marks less than 80:

```
> db.students.find({ marks: { $lt: 80 } })
< {
  _id: ObjectId('6ab001ce167b65c1cd41f652'),
  rollNo: 103,
  name: 'Amit',
  age: 19,
  department: 'ME',
  marks: 75
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f654'),
  rollNo: 105,
  name: 'Rohit',
  age: 21,
  department: 'CSE',
  marks: 78
}
```

D. \$lte – Marks less than or equal to 80:

E. \$ne – Marks not equal to 80:

```
> db.students.find({ marks: { $ne: 80 } })
< {
  _id: ObjectId('6ab0008d167b65c1cd41f650'),
  rollNo: 101,
  name: 'Sejal',
  age: 20,
  department: 'CSE',
  marks: 85
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f651'),
  rollNo: 102,
  name: 'Shruti',
  age: 21,
  department: 'ECE',
  marks: 90
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f652'),
  rollNo: 103,
  name: 'Amit',
  age: 19,
  department: 'ME',
  marks: 75
}
```

5. Logical Operators

Logical operators include \$and, \$or, \$not and \$nor.

A. \$and – Both conditions must be true:

```
> db.students.find({
  $and: [
    { marks: { $gt: 80 } }, { age: { $lt: 21 } }
  ]
})
< {
  _id: ObjectId('6ab0008d167b65c1cd41f650'),
  rollNo: 101,
  name: 'Sejal',
  age: 20,
  department: 'CSE',
  marks: 85
}
```

B. \$or – At least one condition must be true:

```

> db.students.find({
  $or: [
    { department: "CSE" }, { department: "ECE" }
  ]
})
< {
  _id: ObjectId('6ab0008d167b65c1cd41f650'),
  rollNo: 101,
  name: 'Sejal',
  age: 20,
  department: 'CSE',
  marks: 85
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f651'),
  rollNo: 102,
  name: 'Shruti',
  age: 21,
  department: 'ECE',
  marks: 90
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f653'),
  rollNo: 104,

```

C. Find students whose marks are NOT greater than 80.

```

> db.students.find({
  marks: { $not: { $gt: 80 } }
})
< {
  _id: ObjectId('6ab001ce167b65c1cd41f652'),
  rollNo: 103,
  name: 'Amit',
  age: 19,
  department: 'ME',
  marks: 75
}
{
  _id: ObjectId('6ab001ce167b65c1cd41f654'),
  rollNo: 105,
  name: 'Rohit',
  age: 21,
  department: 'CSE',
  marks: 78
}

```

6. Updating Documents

Update one document:

7. Deleting Documents

A. Delete one document:

B. Delete multiple documents:

